Team 4



Project Management Plan

Terrapin Events

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1. Introduction

1.1. Document Purpose

The purpose of this **Project Management Plan (PMP)** is to define the planning, execution, and oversight approach for the development of the Campus Event Management System (**CEMS**) called **Terrapin Events**. This document establishes the scope, objectives, roles, schedules, risks, and progress tracking mechanisms that will guide the project team throughout the Software Development Lifecycle (**SDLC**).

The **Terrapin Events** will provide a centralized digital platform to streamline the planning, registration, and management of campus events at the University of Maryland, College Park. By replacing the existing manual and fragmented processes, the system will improve coordination, reduce administrative overhead, and enhance student and faculty engagement.

This Project Management Plan ensures that project activities are aligned with stakeholder needs and organizational goals, and it serves as a baseline reference for communication, accountability, and decision-making across the project team.

1.2 Project Description

Context

The purpose of **Terrapin Events** is to provide a centralized digital platform for exploring, planning, organizing, registering, and tracking campus events at the **University of Maryland, College Park**. The system will replace outdated manual and fragmented processes with a unified solution that enhances accessibility, efficiency, and engagement.

Objectives

By consolidating event information into a single, accessible platform **Terrapin Events** will:

- Increase awareness of academic, cultural, and extracurricular events across the campus community.
- Streamline event creation, approval, and coordination for organizers while reducing administrative overhead.
- Enhance student and faculty engagement by making event discovery and registration seamless.
- Provide administrators with tools for attendance tracking, reporting, and data-driven decision-making.

Significance

This system is significant because it addresses the inefficiencies of current event management at the university, where processes are manual, fragmented, and prone to low visibility. The lack of a unified platform often results in missed opportunities for engagement and higher administrative overhead. By digitizing the event lifecycle, **Terrapin Events** reduces barriers to participation, strengthens communication, and fosters a more connected campus environment.

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1.2. Alignment with Organizational Goals

Terrapin Events directly supports the University of Maryland's strategic objectives of promoting campus engagement, fostering collaboration, and advancing digital transformation. By enabling equitable access to event information and integrating secure payment processing for paid events, the system enhances the overall student and faculty experience, while aligning with the university's commitment to innovation, inclusivity, and operational efficiency.

Stakeholder-to-Feature Mapping (Extra Credit Attempted)

Stakeholder	Features / Capabilities
University of Maryland (represented by Professor Barberas)	Strategic oversight of the platform to ensure it aligns with institutional goals, compliance with university policies, faculty/student engagement, and long-term sustainability of the event management system.
Students and Faculty	Event discovery via centralized catalog, simple registration, waitlist management, reminders, and notifications.
Event Organizers	Event creation and approval workflows, secure payment gateway for ticketed events, participant communication, and reporting tools.
Administrators	Oversight dashboards, attendance and engagement analytics, event approval workflows, and compliance monitoring.
IT and Security Teams	Integration with UMD CAS, data privacy controls, PCI DSS and FERPA compliance, system monitoring and maintenance.

Table 1: Stakeholders and Feature Mapping

2. Project Team

2.1 Team Members Details

Team Member	Role	Responsibilities
Heena Khan	Project Manager / Software Engineer	Oversee project planning, coordination, and progress tracking. Ensure deliverables meet quality standards. Facilitate communication between team and stakeholders. Support coding, integration, and testing as needed.
Bhavna Kumari	Requirements Engineer	Lead requirements elicitation, documentation, and validation. Collaborate with stakeholders to clarify needs. Ensure

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		requirements are consistent, testable, and aligned with project goals.
Teja Tatimatla	Software Architect	Define overall system architecture and technology stack. Create design diagrams and standards. Ensure scalability, maintainability, and integration between modules. Provide guidance during development.
Vineet Agarwal	Developer	Implement assigned modules according to design. Write clean, efficient, and testable code. Collaborate with the architect and other developers for integration. Assist in debugging and testing.
Abhigna Kandala	Developer	Develop assigned features and modules. Participate in code reviews and testing. Ensure code follows standards and integrates smoothly. Support deployment and maintenance.

Table 2: Team Member Descriptions

2.2 Organisation Chart

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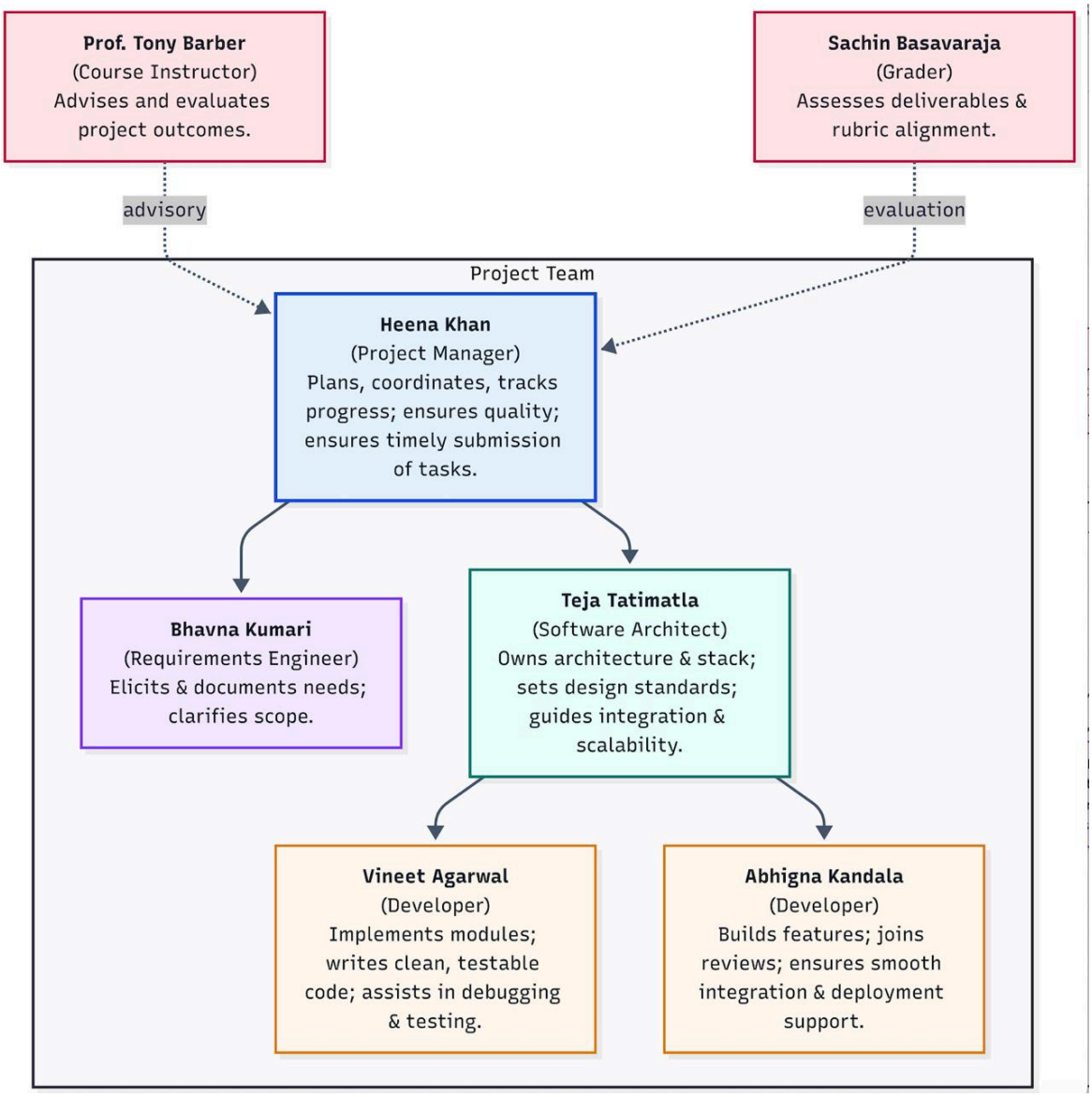


Figure 1: Organisation Chart

3. Project Scope

3.1. Project Boundaries

Boundary Definition

The Boundary of **Terrapin Events** is to provide a centralized digital platform for exploring, planning, organizing, registering, and tracking campus events at the **University of Maryland, College Park**. The system will replace outdated manual and fragmented processes with a unified solution that enhances accessibility, efficiency, and engagement.

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In-Scope

The project will include the following activities and features:

- Requirements gathering, analysis, and documentation for CEMS.
- System design and architecture.
- Development of a web-based application integrated with UMD CAS for authentication.
- Event creation, approval workflows, registration, and waitlist management.
- Calendar integration (e.g., iCal feeds) to sync events with personal and campus-wide calendars.
- Integration of a secure payment gateway to support paid events.
- Notifications and reminders for participants.
- Attendance tracking and analytics dashboards.
- User training and documentation.
- Testing (unit, integration, system, and user acceptance).
- Deployment of the system within the university environment.

Out-of-Scope

The following activities are excluded from this project:

- Development of a native mobile application (may be considered in future phases).
- Integration with non-university financial systems beyond the selected payment gateway.
- Long-term post-deployment maintenance and support beyond the project timeline.
- Management of events hosted outside the University of Maryland ecosystem.

3.2. Identification of Assumptions and Constraints

Assumptions

- All users (students, faculty, staff, and administrators) will have a valid **UMD Directory ID** and access to UMD CAS for authentication.
- The group members will provide necessary infrastructure and hosting support (servers, network, and IT security compliance).
- A single **secure payment gateway** will be selected, approved, and supported by the university's financial/IT policies.
- Users will have access to internet-connected devices (laptops, tablets, smartphones) and modern browsers to interact with the system.
- Stakeholders (students, faculty, administrators) will actively participate in requirements elicitation, testing, and feedback cycles.
- The project will be completed within the allocated timeline and resource availability as defined in the plan.

Constraints

- The system must comply with **University of Maryland IT security policies** and industry standards (e.g., PCI DSS for payment transactions, FERPA for student data).

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- The scope of development is limited to a **web-based application**; a mobile-native app will not be implemented in this phase.
- Only **University of Maryland–affiliated events** will be managed; external or community events are excluded.
- The project is constrained by **time and academic semester deadlines**, requiring phased delivery of features.
- Post-deployment **long-term maintenance and operational support** are outside the scope of this project.
- Integration with third-party services will be limited to university-approved and secure platforms, such as the selected payment gateway. Unsupported external ticketing or unapproved payment systems will not be integrated.

3.3. Linkage between business needs and project objectives

Background and Problem Statement

Currently, event planning and promotion at the University of Maryland rely on disparate tools such as email announcements, flyers, and independent department websites. These fragmented methods create gaps in communication and reduce awareness, leading to under-attended events and inefficient use of resources. Organizers also face challenges in managing approvals, tracking registrations, and handling payments for ticketed events.

Intended Benefits

Terrapin Events will provide the following benefits:

- **For Students and Faculty**: Simplified access to a comprehensive event catalog, easy registration, and improved awareness of campus opportunities.
- **For Organizers**: Streamlined workflows for event creation, approval, and payment handling, reducing duplication of effort.
- **For Administrators**: Robust attendance tracking, reporting tools, and analytics to support better planning and resource allocation.

Alignment with Organizational Goals

Terrapin Events directly supports the University of Maryland's strategic objectives of promoting campus engagement, fostering collaboration, and advancing digital transformation. By enabling equitable access to event information and integrating secure payment processing for paid events, the system enhances the overall student and faculty experience, while aligning with the university's commitment to innovation, inclusivity, and operational efficiency.

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Work Context Diagram (Extra Credit Attempted)

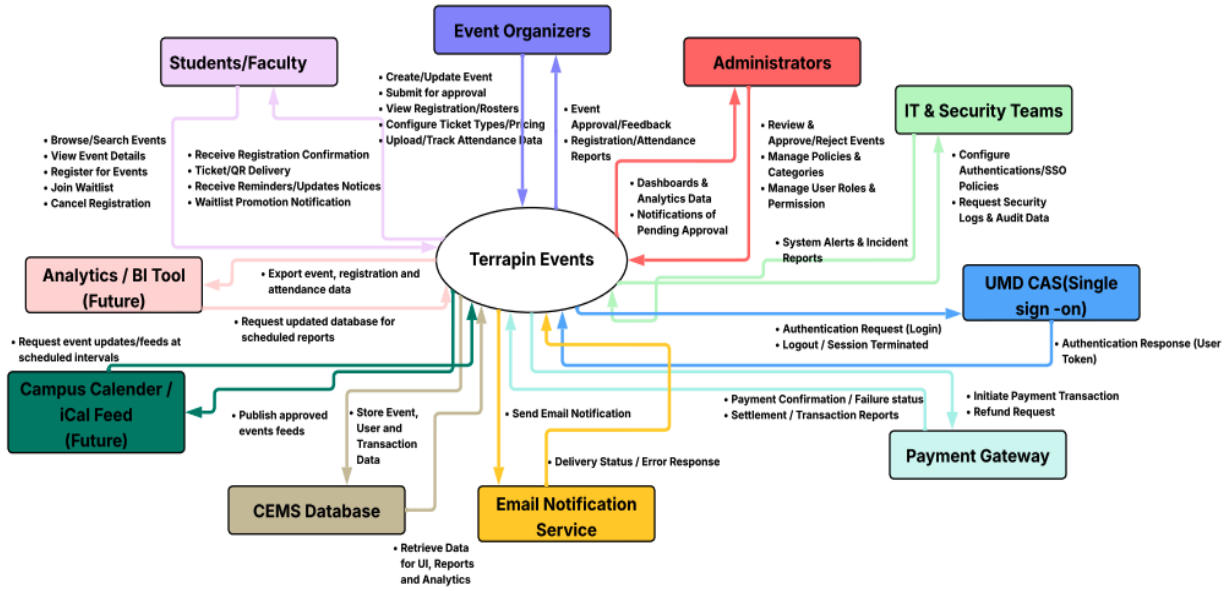


Figure 2: Work Context Diagram

4. Project Concept

The **Terrapin Events** system establishes the foundation for a modernized approach to event management at the University of Maryland, College Park. This section articulates stakeholder business needs, defines key functional and non-functional requirements, and provides a high-level description of the planned features and system capabilities that will guide the design and implementation of the software solution.

4.1. Articulation of stakeholder business needs

- **Students and Faculty:** Need a single, user-friendly platform to explore, register for, and stay updated on campus events, ensuring greater accessibility and participation.
- **Event Organizers:** Require tools to efficiently create, promote, and manage events, including approval workflows, secure ticket sales for paid events, and participant communications.
- **Administrators:** Need oversight of event activities through analytics, attendance tracking, and reporting to support informed decision-making and ensure policy compliance.
- **IT and Security Teams:** Require seamless integration with UMD CAS authentication, adherence to university IT policies, and compliance with industry standards for data privacy and payment security.

4.2. Identification of functional and non-functional requirements

Functional Requirements (High-Level):

The **Terrapin Events** system must provide the following functional capabilities:

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- **FR001- Authentication & Access Control** – All users shall log in using UMD CAS, with role-based access granted to students, faculty, event organizers, administrators, and IT staff.
- **FR002- Event Creation & Approval** – Event organizers shall be able to create events with relevant details (title, description, category, date/time, venue, and capacity), subject to an approval workflow managed by administrators.
- **FR003- Registration & Waitlist Management** – Users shall be able to register for events, view confirmation details, and be placed on automated waitlists when capacity is reached.
- **FR004- Payment Gateway Integration** – The system shall integrate a secure payment gateway to support ticketed or premium events, allowing participants to complete transactions online.
- **FR005- Notifications & Reminders** – The system shall send automated event reminders, registration confirmations, and updates (e.g., cancellations or changes) via email and/or in-app notifications.
- **FR006- Attendance Tracking & Reporting** – Event organizers and administrators shall be able to track attendance, generate reports, and access analytics dashboards to evaluate participation trends.
- **FR007- Search & Discovery** – Students and faculty shall have access to a centralized event catalog with search and filter capabilities (e.g., by date, type, or organizer).
- **FR008- Post-Event Feedback -** The system shall allow participants to submit feedback after attending an event, and provide organizers and administrators access to feedback summaries.
- **FR009- Calendar Integration** - The system shall allow users to add registered events to their personal calendars (e.g., Google Calendar, Outlook, iCal) and sync approved events with the campus-wide calendar.

Non Functional Requirements (High-Level):

The **Terrapin Events** system must adhere to the following non-functional requirements to ensure quality, security, and usability:

- **NFR001- Reliability & Availability** – The system shall maintain high availability (≥ 99% uptime) to support campus-wide access, particularly during peak registration periods.
- **NFR002 - Security & Compliance** – The system shall comply with **PCI DSS** standards for payment transactions, **FERPA** for student data protection, and all University of Maryland IT security policies.
- **NFR003 - Scalability** – The system shall be designed to handle growth in user load, with the ability to support thousands of concurrent users during major events.
- **NFR004 - Performance** – Event searches, registrations, and updates shall process within 2 seconds under normal load to ensure responsiveness.
- **NFR005 - Usability & Accessibility** – The user interface shall be intuitive, ADA-compliant, and compatible with modern browsers across devices (desktop, tablet, mobile).
- **NFR006 - Maintainability** – The system shall be modular and documented to facilitate ongoing updates, bug fixes, and enhancements.
- **NFR007 - Data Integrity & Backup** – The system shall ensure secure storage of event and user data, with automated daily backups and disaster recovery mechanisms in place.

4.3. High Level Description of Planned Features and System Capabilities

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The **Terrapin Events** system will provide the following high-level features and capabilities:

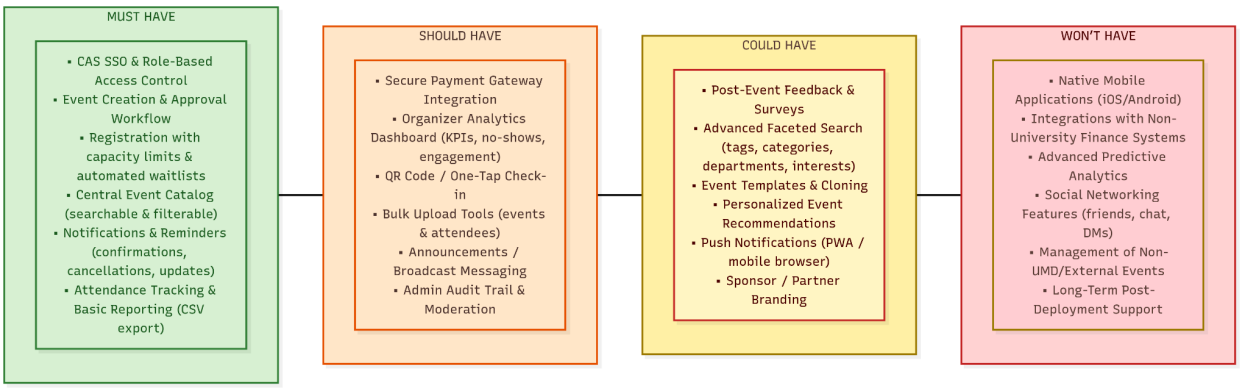
- **Centralized Event Catalog:** A searchable, filterable repository of all campus events.
- **Smart Registration System:** Simple event registration, automated waitlist handling, and real-time confirmations.
- **Secure Payment Gateway:** Online payment processing for ticketed or premium events.
- **Analytics Dashboard:** Tools for administrators and organizers to monitor attendance, analyze participation trends, and generate reports.
- **Communication Tools:** Automated notifications, reminders, and event updates to improve engagement.
- **Role-Based Access:** Differentiated access for students, organizers, administrators, and IT staff to ensure proper security and accountability.

Prioritised Capability List (Extra Credit Attempted)

The Prioritised Capability List defines both the functional and non-functional capabilities of the **Terrapin Events** system, ranked by importance using the MoSCoW method (Must Have, Should Have, Could Have, and Won't Have). This prioritisation ensures that the Minimum Viable Product (MVP) delivers the essential capabilities required for reliable event management, while also identifying enhancements that may be considered in later iterations if resources and time allow.

- **Must Have:** Capabilities that are essential for the system to achieve its intended purpose. These directly enable user authentication, event creation, registration, catalog access, notifications, reporting, and compliance requirements.
- **Should Have:** Capabilities that are important but not mission-critical in the initial release. They provide additional value—such as secure payments, analytics dashboards, and audit trails—but the system remains viable without them.
- **Could Have:** Capabilities that enhance usability and user engagement but are optional within the project scope. These may be implemented if additional resources and time are available.
- **Won't Have:** Capabilities that are explicitly excluded from the current release. These are documented to manage stakeholder expectations and prevent scope creep.

This prioritisation approach helps balance **functionality, schedule, and resources**, ensuring that student and faculty needs are met first, while also providing a roadmap for future development phases.



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Figure 3: MOSCOW Method of Prioritization

5. Work Breakdown Structure (WBS)

Our Work Breakdown Structure (WBS) was designed to focus on major phases and deliverables rather than an exhaustive task list. This keeps the project manageable, transparent, and outcome-oriented, ensuring that each stage produces tangible artifacts such as specifications, diagrams, or prototypes. By assigning a responsible role to each phase, we maintain accountability while still encouraging collaboration across the team.

While the WBS presents a traditional phase-based breakdown, our team is executing the project in six sprints, each aligned to a major deliverable. This Agile-inspired approach allows us to start tasks early, overlap work when feasible, and incorporate continuous feedback cycles. For example, UI prototyping begins while requirements are being refined, enabling early validation without disrupting dependency flows.

5.1. Logical decomposition of project into phases, tasks, and deliverables

WBS ID	Phase/Sprint	Deliverables	Description	Responsible Role
1.0	Project Management Plan (Sprint 1)	PMP, WBS, RACI, Risk Register, Progress Tracker	Develop and refine the WBS, responsibility matrix (RACI), risk register, and consolidated schedule.	Project Manager
2.0	Requirements (Sprint 2)	Requirements Spec, Abuse Cases, RTM	Document functional and non-functional requirements, abuse cases, and construct the traceability matrix.	Requirements Engineer
3.0	User Model & UI Prototype (Sprint 3)	Use Case Diagram, Narratives, Prototype	Create UML use case diagrams and create a low- to medium-fidelity user interface prototype.	Requirements Engineer + Developer
4.0	Software Architecture (Sprint 4)	Architecture Doc, Component/Deployment/Sequence Diagrams	Develop system architecture documentation including component and deployment diagrams with rationale.	Software Architect
5.0	Detailed Design (Sprint 5)	Detailed Design Doc, Class/Activity/State Diagrams	Refine system design into modules and classes, generate UML class/activity/state diagrams.	Architect + Developer
6.0	Initial Implementation	Prototype	Begin coding core modules and integrations.	Developers

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	(Sprint 6)			
7.0	Continued Development and Testing (Sprint 7)	Test Plan, Test Cases, Coverage Report	Expand features and refine prototypes based on testing feedback.	QA + Developers
8.0	Deployment and Final Presentation (Sprint 8)	Final Deck, Demo, Integrated Repo	Deploy software to the production environment and deliver the final project presentation.	Project Manager + Entire Team

Table 3: Work Breakdown Structure

6. Task Allocation & Responsibility Matrix (RACI)

The **RACI** Matrix is used to define and clarify the level of involvement each team member has for major project tasks. It ensures accountability, prevents role confusion, and aligns responsibilities with each member's expertise.

- **R – Responsible:** The person(s) who performs the task or deliverable. They are directly responsible for execution.
- **A – Accountable:** The single person who is ultimately answerable for the completion of the task. They approve work and ensure it meets objectives.
- **C – Consulted:** Individuals whose input is sought to provide expertise, guidance, or feedback.
- **I – Informed:** Individuals who must be kept up to date on progress and outcomes but are not directly involved in execution.



Figure 4: RACI Description

In this project, the RACI matrix assigns roles across all major phases of the Software Development Lifecycle (SDLC), such as requirements gathering, system design, development, testing, deployment, and

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risk management. By mapping tasks to Responsible, Accountable, Consulted, and Informed roles, the matrix ensures that every activity has clear ownership, balanced workload distribution, and transparent communication channels.

6.1. Inclusion of major project tasks with R, A, C, I roles defined

Task/Deliverable	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements Gathering	Requirements Engineer	Project Manager	Software Architect	All Team Members
Software Architecture	Software Architect	Project Manager	Developers	All Team Members
Module Development	Developers	Project Manager	Software Architect	Entire Team
System Testing	Developers (Testers)	Project Manager	Developers	All Team Members
Deployment	Developers	Project Manager	Stakeholders	All Stakeholders

Table 4: Task Allocation & Responsibility Matrix

6.2. Balanced Allocation of Responsibilities

Responsibilities in this project are distributed across the team to ensure that every member contributes meaningfully according to their role and expertise, while avoiding dependency on a single individual. The Project Manager oversees planning, coordination, and approvals; the Requirements Engineer focuses on stakeholder engagement and requirement specification; the Solution Architect leads system design and ensures technical feasibility; and the Developers share implementation, testing, and deployment responsibilities.

This balanced allocation ensures that:

- No single member is overloaded with multiple critical tasks.
- Each task has both **responsible executors** and an **accountable owner** for quality assurance.
- Cross-consultation (via the RACI model) ensures collaboration without duplication of effort.
- Workload is distributed across the full Software Development Lifecycle (requirements → design → development → testing → deployment).

As a result, all members contribute equitably, expertise is leveraged effectively, and the team maintains flexibility to cover risks such as absences or unexpected challenges.

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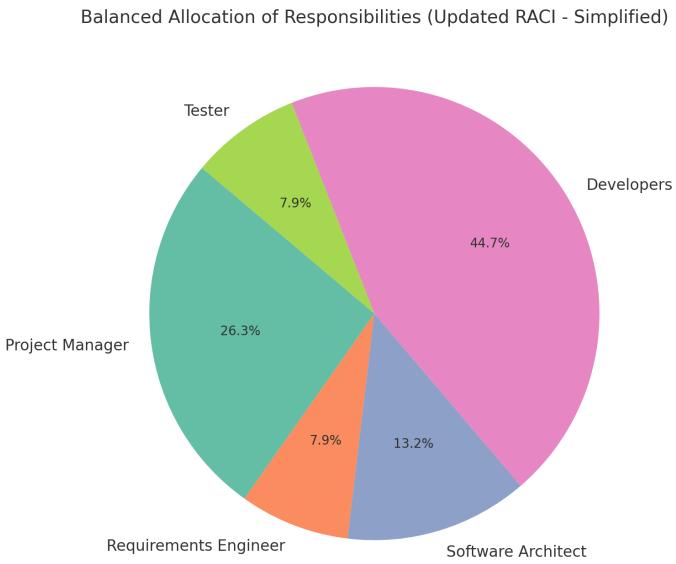


Figure 5: Balanced Allocation of Responsibilities

6.3. Evidence of alignment between RACI and WBS

WBS ID	Deliverable (WBS)	RACI Assignment
1.0	Project Management Plan (PMP, WBS, RACI, Risk Register, Progress Tracker)	Responsible (R): Project ManagerAccountable (A): Project ManagerConsulted (C): Entire TeamInformed (I): Stakeholders
2.0	Requirements Specification, Abuse Cases, RTM	(R): Requirements Engineer (A): Project Manager (C): Software Architect (I): Entire Team
3.0	User Model & UI Prototype (Use Case Diagrams, Narratives, Prototype)	(R): Requirements Engineer + Developers (A) : Project Manager (C): Architect (I): Entire Team
4.0	Software Architecture (Architecture Doc, Component/Deployment/Sequence Diagrams)	(R): Software Architect (A): Project Manager (C): Developers (I): Entire Team
5.0	Detailed Design (Design Docs, Class/Activity/State Diagrams)	(R): Developers (A): Project Manager (C): Software Architect (I): Entire Team
6.0	Initial Implementation (Prototype, Core Module Coding)	(R): Developers (A): Project Manager (C): Software Architect (I): Entire Team

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7.0	Continued Development & Testing (Test Plans, Test Cases, Coverage Report)	(R): Tester (A): Project Manager (C): Developers (I): Entire Team
8.0	Deployment & Final Presentation (Demo, Integrated Repo, Final Deck)	(R): Developers (A): Project Manager (C): Faculty Advisor (I): All Stakeholders

Table 5: Alignment between RACI and WBS

7. Project Schedule & Timelines

This section decomposes the Terrapin Events project into discrete work packages. Each work package represents a well-defined development activity or milestone mapped to the Agile sprint cycle. Sprints are planned as one- or two-week intervals, allowing for incremental delivery and review.

Agile Overview (Extra Credit Attempted)

Agile is an iterative approach to software development that promotes flexibility, collaboration, and customer-centric delivery. It breaks the project into manageable increments called sprints, enabling the team to adapt quickly to changes and deliver functional components frequently. Agile emphasizes continuous stakeholder involvement, early detection of issues, and incremental refinement of requirements and design through regular feedback loops.

Agile Methodology Statement (Extra Credit Attempted)

The project will employ the Agile development methodology, characterized by iterative and incremental delivery through bi-weekly sprints. Each sprint will begin with planning and task prioritization, followed by daily standup meetings and routine progress evaluations. Reviews will occur at the end of each sprint to assess deliverable quality and elicit stakeholder feedback, ensuring continuous alignment with goals and requirements. Adaptations to the project plan will be made based on feedback from retrospective meetings.

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Team 4 Jira Board : [Jira Board team 4]

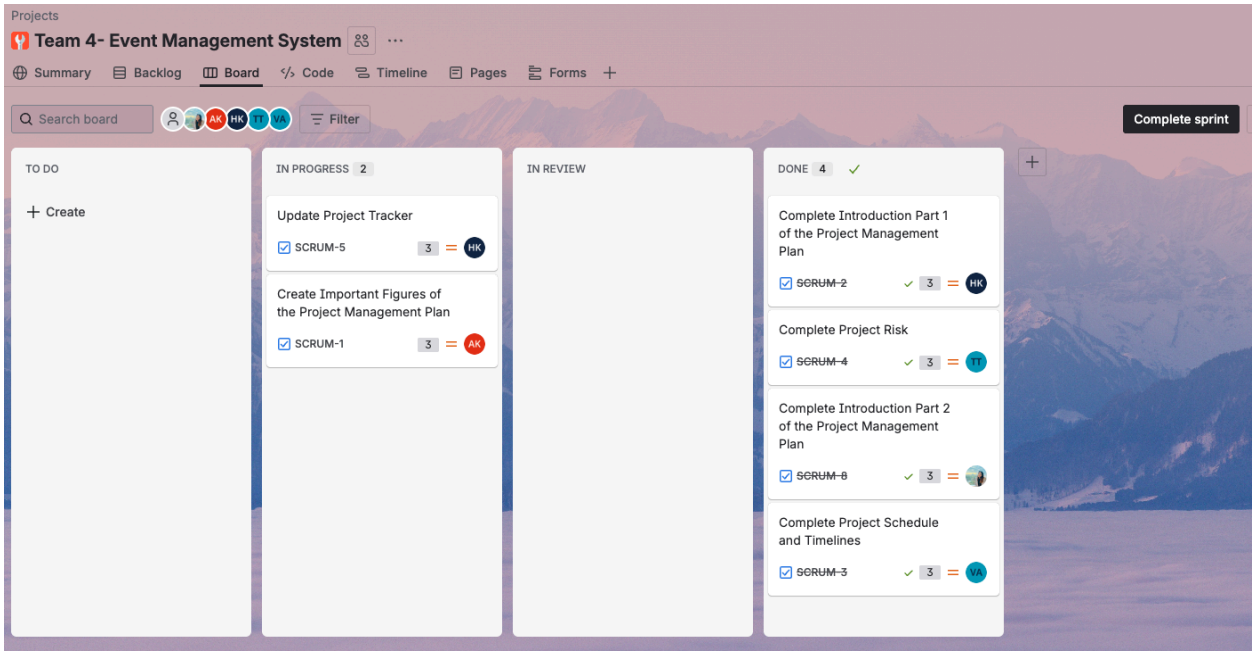


Figure 6: Sprint 1 - Project Management Plan

Planned Project Schedule (Extra Credit Attempted)

The project schedule is monitored and managed using Jira boards, supporting sprint planning, task tracking, and reporting. The schedule includes contingency buffers to manage risks of delays or resource unavailability. We intentionally adopted a **staggered schedule** where each sprint begins approximately **one week ahead** of the official deadline. This creates **buffer periods** for internal reviews, quality checks, and risk mitigation, reducing the likelihood of last-minute issues. Dependencies remain respected—for example, architecture only builds on stabilized requirements, but the overlap maximizes productivity and ensures we comfortably conclude by the **final milestone on November 25**.

7.1. Identification of key milestones

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Milestone	Description	Planned Start Date	Planned Completion Date	Effort Estimate(Weeks)
Project Management Plan	PMP finalized with WBS, RACI, risks, and initial Progress Tracker submitted	9-18-25	10-3-25	2
Requirements Complete	Requirements Specification, Abuse Cases, RTM finalized and approved	9-27-25	10-17-25	3
User Model & UI Prototype	UML Use Case Diagrams, Narratives, and UI Prototype developed and reviewed	10-10-25	10-24-25	2
Architecture Complete	Software Architecture Document with Component, Deployment, and Sequence Diagrams completed	10-18-25	11-7-25	3
Detailed Design Complete	Detailed Design Document with Class, Activity, and State Diagrams finalized	11-1-25	11-15-25	2
Implementation & Testing	Minimal Working Prototype built; Test Plan, Test Cases, and Coverage Reports delivered	11-9-25	11-23-25	2
Deployment and Final Presentation	Release software to production/demo environment and conduct final project presentation.	11-17-25	11-25-25	1

Table 6: Key Milestone Identifier

7.2. Logical sequencing of activities with estimated timelines

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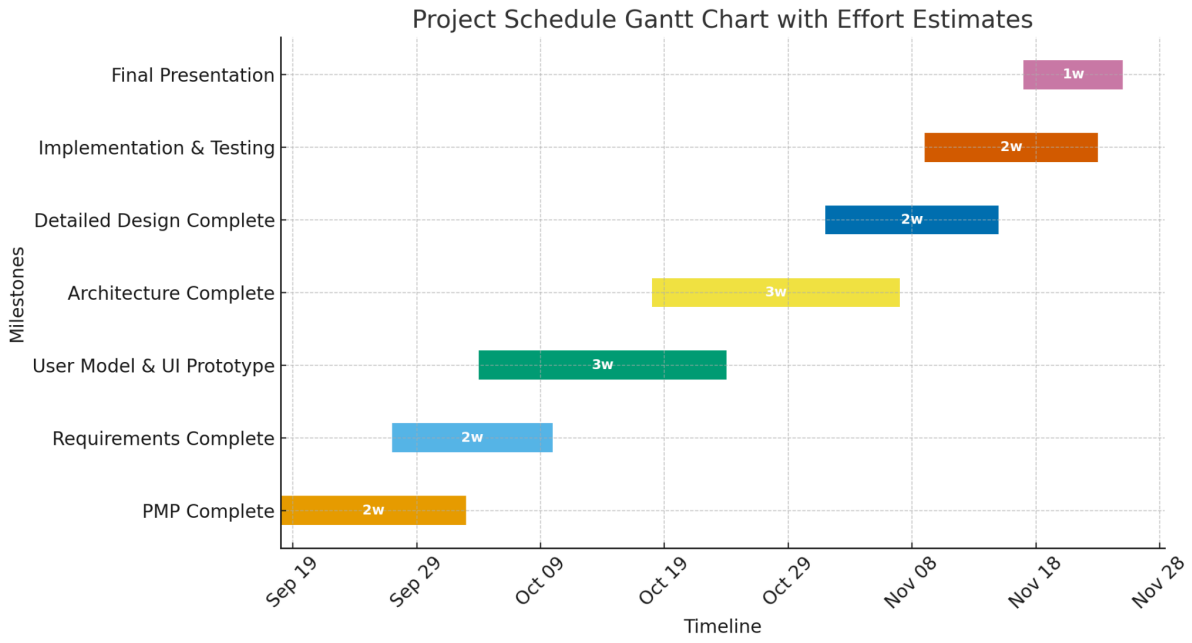


Figure 7: Project Schedule (Planned)

7.3. Realistic Scheduling Demonstrating Planning Rigor

The sprint-based schedule demonstrates planning rigor by sequencing tasks in alignment with the SDLC. Requirements are completed before design, design precedes development, and testing is scheduled only after modules are implemented. Each sprint has achievable deliverables aligned with WBS and RACI roles, ensuring realistic workload distribution. This structured breakdown reduces risk of bottlenecks and provides clear checkpoints for tracking progress.

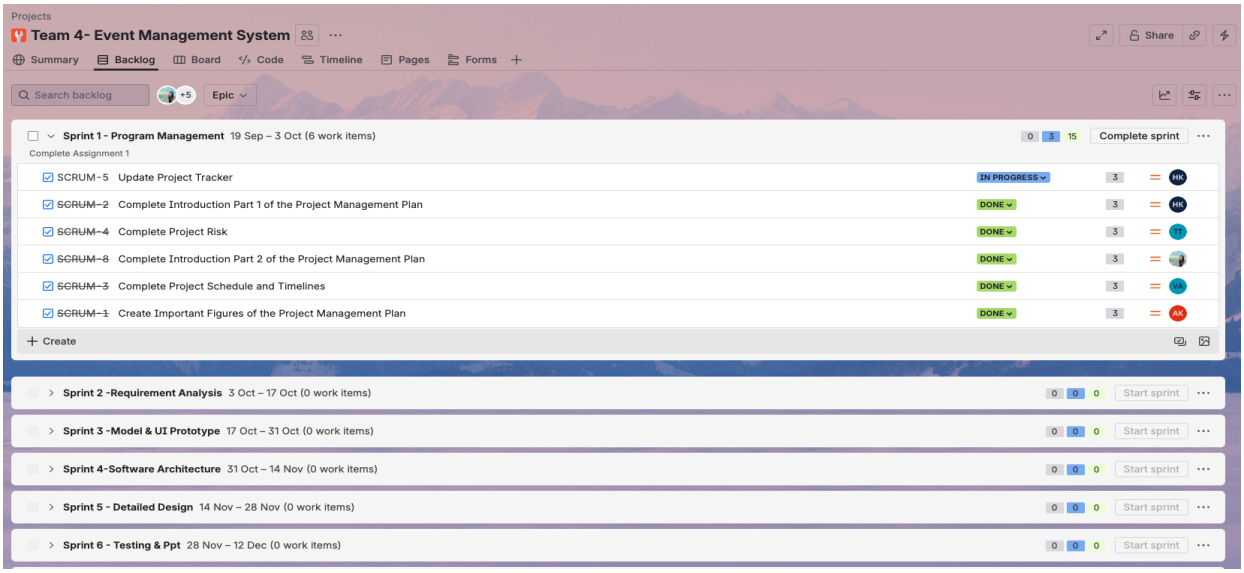


Figure 8: Project Schedule Backlog (Onsite)

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8. Project Risk

8.1. Risk Approach Overview

This section describes risks specific to developing and operating the **Terrapin Events** application. Each risk is assessed for **Likelihood** and **Impact** and includes a **Mitigation/Response** plan and an **Owner** responsible for monitoring and initiating mitigation. The team will review this register at each sprint review and reevaluate the **Likelihood** and **Impact**.

Assumptions And Constraints

- Part-time workforce with study and assignments.
- Student and faculty data is subject to FERPA and university IT policies.
- UI/UX must follow the university's style guide.
- Integrations with UMD CAS, email, and payment gateways.
- Target browsers are Chrome/Firefox/Edge/Safari, mobile usage is not expected

Risk ID	Description	Likelihood (Low/Med/ High)	Impact (Low/Med/ High)	Mitigation/Response	Owner
R-1	Underestimated effort or resource availability issues due to competing coursework or illness	Med	High	Planning sprints with buffer, escalating blockers within 24hrs.	Heena
R-2	Scope creep through new feature requests	Med	Med	PM approval, prioritizing backlog, limiting scope changes after MVP	Bhavna
R-3	CAS integration failure	Med	High	Start CAS spike in Sprint 1, work with campus IT	Vineet/ Abhigna
R-4	FERPA non-compliance	Low	High	Role-based access control, least privileges, audit logs for admin accounts, privacy review checklist, work with campus IT	Teja
R-5	Web application vulnerabilities	Med	High	input/output encoding, dependency scanning, OWASP top ten checklist, pre-release pen-testing	Teja
R-6	Accessibility gaps: WCAG 2.1 AA compliance	Med	Med	Semantic HTML, keyboard nav, color-contrast test, screen-reader test	Abhiga/ Vineet

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R-7	Peak-load performance degradation	Med	High	Define acceptable response times, load testing, caching, scale plan	Teja
R-8	Accidental data loss	Low	High	Daily automated backups, implementing point-in-time recovery	Teja
R-9	Email notifications landing in spam	Med	Med	Using reputable SMTP provider or use university's SMTP	Abhigna /Vineet
R-10	Third-party API dependencies risks (rate limits, etc.,)	Low	Med	Retries with backoff, implementing controls to disable non-critical integrations gracefully	Teja
R-11	Usability/Adoption risk	Med	Med	Early usability tests with organizers, simple onboarding, quick-start guide	Heena/ Bhavna
R-12	Payment data handling compliance	Low	High	Avoid storing credit/debit data, only use well established payment gateways, data logs	Heena/ Bhavna
R-13	Campus policy misalignment	Low	Med	Review IT policy checklist, work with campus IT	Heena
R-14	Browser compatibility issues	Med	Med	Smoke tests on target browsers, capture client error and fix-high volume issues first	Bhavna

Table 7: Risk Mitigation Plan

Monitoring And Review

Review the risk register weekly and update Likelihood and Impact after sprint review or after any incident. Any high impact and med/high likelihood risk should trigger an immediate mitigation plan following PM escalation to primary stakeholders.

System of Interest (SOI) (Extra Credit Attempted)

This project is a centralized digital platform designed to streamline the planning, organization, registration, and tracking of campus events at the University of Maryland. It provides a unified interface for students, faculty, and administrators to create, manage, and participate in events efficiently. The system includes features such as event scheduling, venue allocation, participant registration, and notification services. **The SOI emphasizes scalability, usability, and security, while ensuring smooth communication and collaboration across the campus community.**

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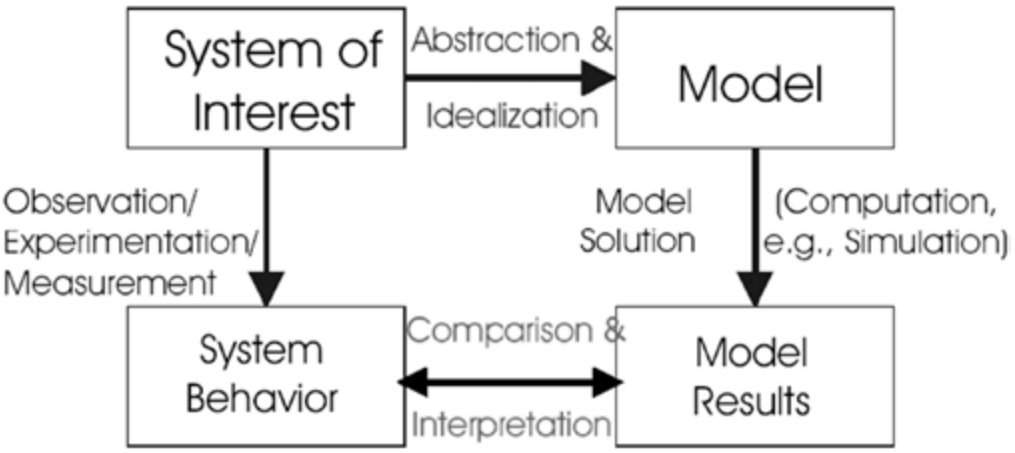


Figure 9: System of Interest

Measures of Effectiveness (MOEs) (Extra Credit Attempted)

MOE Category	Measure of Effectiveness (MOE)	Target / Threshold
System Availability	Percentage uptime during academic year	≥ 99% uptime
Performance	Average page load/transaction response time	≤ 2 seconds for 95% of requests
Scalability	Maximum concurrent users supported during peak registration	At least 5,000 concurrent users
Accessibility	Compliance with accessibility standards	WCAG 2.1 AA & ADA compliant
Security	Data privacy & regulatory compliance	0 critical FERPA violations; pass annual security audit
User Adoption	Percentage of student population using system per semester	≥ 70% of students
Engagement	Percentage of events created & managed via CEMS vs. legacy/manual methods	≥ 85% of campus events
User Satisfaction	Positive responses in post-event/portal feedback surveys	≥ 80% satisfaction rate

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Reliability	Number of system crashes or data losses during semester	≤ 2 incidents per semester

Table 8: Measures of Effectiveness

9. Progress Tracking & Reporting

9.1. Progress Tracker

In addition to the Progress Tracker (Progress Tracker.xlsx), our team is using Jira to manage tasks, monitor sprint progress, and provide real-time visibility into deliverable status.

Jira dashboards and burndown charts will supplement weekly progress updates to ensure accountability and transparency.

Jira Board : Team 4 Jira Board

Project Tracker Sheet : Link

Progress Tracker

Effort (Person hours)

MAIN ACTIVITY (Use the WBS to Update)	Sub-tasks	Heena Khan (Project Manager)		Bhavna Kumari (Requirements Engineer)		Teja Tatimatla (Solutions Architect)		Vineet Agarwal (Developer)		Abhigna Kandala (Developer)		TOTAL ESTIMATED EFFORT per activity	TOTAL ACTUAL EFFORT per activity
	Sub-tasks	Estimated	Actual	Estimated	Actual	Estimated	Actual	Estimated	Actual	Estimated	Actual		
Project Management Plan	Draft PMP (Structure, purpose, Context)			8	7							8	7
	Create Work Breakdown Strcuture (WBS)							8	7			8	7
	Develop RACI Matrix	6	6									6	6
	Risk Register Preparation					8	8					8	8
	Preparation of Figures									8	8	8	8
	Progress Tracker SetUp	2	2									2	2
	Review & Finalize PMP Document	2	2	2	2	2	2	2	2	2	2	10	10
Requirements	Conduct stakeholder interviews & surveys												
	Document functional requirements												
	Document non-functional requirements and design constraints												
	Build requirements traceability matrix (RTM)												
	Validate requirements with stakeholders												
	Analysis												
Design	Ux design											0	0
	Architecture design											0	0
	Database design											0	0
	Architecture documentation (SAD)											0	0
	Detailed design (SADD)											0	0
Implementation	Setup											0	0
	Module development											0	0
	Integration											0	0
	Code quality											0	0

Figure 10: Effort Estimation and Tracking

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Appendix A

References

- Grading Rubric provided by the professor
- Project Management Plan slides provided by the professor
- Project Management Plan Template provided by the professor
- Jira by Atlassian
- System of Interest Inspiration taken from the research Paper → ***Simulation, Models, and Results: Reflections on their Nature and Credibility***
- Work Breakdown Structure made on Lucid Chart
- Project Organisation Chart made on Mermaid Chart
- Project Management Plan Image taken from flaticon.com

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Appendix B

Acronyms

- **UMD CAS (Central Authentication Service):** A single sign-on protocol used by the University of Maryland to authenticate students, faculty, and staff. All users must have a UMD Directory ID to access the system.
- **FERPA (Family Educational Rights and Privacy Act):** A federal law that protects the privacy of student education records; relevant for data security in event participation records.
- **PCI DSS (Payment Card Industry Data Security Standard):** A global security standard for handling credit card transactions securely.
- **SOI(System of Interest):**The specific system being developed or studied that is the primary focus of the project and its requirements.
- **MOE(Measure of Effectiveness):**The specific system being developed or studied that is the primary focus of the project and its requirements.
- **WBS (Work Breakdown Structure):**A logical decomposition of the project into phases, tasks, and deliverables.
- **RTM(Requirements Traceability Matrix):** A document mapping each requirement to its corresponding design, implementation, and test cases to ensure full coverage.
- **PMP(Project Management Plan):** The document that outlines the planning, execution, monitoring, risk and control strategy of the project.
- **RACI(Responsible Accountable Consulted Informed):** A responsibility assignment matrix clarifying team roles for each task/deliverable.
- **QA(Quality Assurance):** The systematic process of ensuring that deliverables meet quality standards via reviews, testing, and compliance checks.

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Appendix C

Definitions

- **Event Registration:** The process by which students and faculty reserve a spot in an event; capped by event capacity with waitlist functionality.
- **Payment Gateway:** A secure online service for processing event payments. The gateway must comply with PCI DSS standards to protect sensitive payment information.
- **Event Analytics:** The process of collecting and analyzing participation data (e.g., attendance rates, no-shows) to support university planning and decision-making.
- **ADA Compliance:** Digital accessibility standards (Section 508) requiring that software be usable by individuals with disabilities (e.g., screen readers, keyboard navigation).
- **Waitlist Management:** A queue system that automatically assigns participants to open slots when existing registrants cancel or fail to confirm.

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